
Merging Specialist Diffusion Models for Multi-Task Astronomical Image Restoration

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Abstract

Astronomical image-processing pipelines typically chain several independent restoration steps, each best solved by a dedicated tool. We investigate whether such a pipeline can be collapsed into a *single* set of weights via post-hoc model merging, avoiding both the cost of serving multiple networks and the need for joint multi-task training data. In particular, we focus on *star removal*, *defect correction* and *super-resolution*. Three *v-prediction* DDPMs are independently fine-tuned (one per task) from a common, purpose-pretrained initialization θ_0 , and their task vectors are combined with Task Arithmetic, TIES-Merging, and DARE-TIES. Crucially, because initial zero-shot merging configurations failed to meet performance expectations due to destructive weight interference, we introduce a brief post-merge fine-tuning stage for all three methods. Merging and fine-tuning hyper-parameters are chosen via a *two-stage grid search* evaluated on all 7 combinations of the three degradations, using manually-curated Hubble Legacy Archive (HLA) imagery. Code is available [here](#).

1. Introduction

Restoring astronomical images often requires multiple sequential corrections, including star removal, detector defect restoration (dead/hot pixels), and super-resolution. A common deep learning approach is to train one specialist model per task, but this requires storing and executing multiple networks.

Model merging combines independently fine-tuned checkpoints into a single model without joint training. While ef-

fective for classification and language models, we find that it performs poorly when applied zero-shot to conditional diffusion models for low-level image restoration. We find that a lightweight post-merge fine-tuning stage helps mitigating task interference and obtain a practical multi-task model.

[TODO: 1–2 sentences summarizing contributions as a list, once results are final, e.g. “(1) we show that X; (2) we find that TIES/DARE-TIES combined with post-merge fine-tuning outperforms naive Task Arithmetic by Y%.”]

2. Related Work

Diffusion models. Denoising Diffusion Probabilistic Models (DDPMs) learn to invert a fixed noising process (Ho et al., 2020). We adopt the *v-prediction* parameterization (Salimans & Ho, 2022), a cosine noise schedule (Nichol & Dhariwal, 2021), and classifier-free-style condition dropout so that the network retains some unconditional generative capacity; the backbone is a conditional U-Net (Ronneberger et al., 2015) with self-attention (Vaswani et al., 2017) at low resolutions.

Model merging. Task Arithmetic (Ilharco et al., 2023) combines task vectors by adding scaled weight differences to a shared pretrained model. TIES-Merging (Yadav et al., 2023) reduces interference by trimming low-magnitude updates and resolving sign conflicts, while DARE (Yu et al., 2024) further sparsifies task vectors through random pruning and rescaling. All methods require checkpoints to share the same pretrained origin, motivating our shared- θ_0 stage (Sec. 3.2).

Prior work targets mainly classifiers and LLMs, and in diffusion mostly LoRA merging for personalization rather than fully fine-tuned specialists on distinct low-level restoration tasks; **[TODO: verify/extend this claim with a short literature search closer to submission – this is the part of the related work most likely to be out of date, and it is also our best candidate for a genuine novelty claim, so it deserves a careful check rather than a**

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hedge.]

3. Method

3.1. Dataset

Source imagery and pre-processing. We start from a manually curated set of color, already-drizzled mosaics from the HLA (Space Telescope Science Institute (STScI), 2006). Raw frames are converted to a working dynamic range via a nonlinear stretch before any degradation is applied. Applying a uniform, global stretching function across the entire dataset yielded highly suboptimal results due to the vast diversity in exposure levels, noise floors, and dynamic ranges among different astronomical targets. To overcome this, an image-specific adaptive arcsinh stretch is performed on each pair before training.

Dataset generation. For all three tasks the supervision target is always the fully clean, stretched image; only the *condition* (network input) differs. Each specialist therefore sees, as condition, an image degraded with only its own corruption, but is only ever supervised to recover the clean image – as opposed to θ_0 's pretext task (Sec. 3.2), where the degradation is deliberately *not* tied to the task the source image came from.

1. **Star removal.** The clean image is processed with StarNetV2 (Misiura, 2020) to obtain a preliminary starless prediction. Rather than using it as ground truth, we use it only to isolate real stars by subtracting it from the original image. The extracted star patches are then re-injected onto pristine starless backgrounds through linear blending, producing controlled input–target training pairs. [TODO: make this definitely shorter]
2. **Dead/hot-pixel restoration.** Defective pixels are identified using a robust outlier detector based on the residual from a local median filter (5×5). A pixel is marked as defective when its channel residual exceeds 6.0σ , where σ is estimated from the Median Absolute Deviation (MAD) scaled by 1.4826. Since HLA RGB composites are assembled from independent filter exposures, defects are sparse and typically affect only a single color channel. [TODO: make this definitely shorter]
3. **Super-resolution.** Paired training data are generated by degrading each high-resolution image with Gaussian blur ($\sigma = 1.5$) followed by $\times 4$ downsampling using cubic interpolation. The resulting low-resolution image is then upsampled back to the original dimensions via bicubic interpolation, providing an aligned

conditioning input for the DDPM. [TODO: make this definitely shorter]

After each of these processes are done, randomized patch sampling and data augmentation strategies extracts sub-images exclusively from regions verified by a geometric validity mask.

[TODO: MANCA SPIEGARE IL DATASET MERGED]

3.2. Shared initialization θ_0

All three specialists are fine-tuned from a common checkpoint θ_0 , obtained by pretraining the same architecture from a random initialization on a *generic conditional-restoration* pretext task, rather than starting each specialist from an independent random init, as required by the original merging methods themselves.

More specifically, θ_0 is trained to reconstruct clean images from synthetically degraded inputs. Degradations consist of 1–3 randomly selected operations from {Gaussian/Poisson-like noise, Gaussian blur, bicubic downsample-upsample, and small local occlusions}, with independently sampled parameters and a 10% probability of a fully zeroed condition (classifier-free guidance dropout (Ho & Salimans, 2022)). This decorrelates image content from degradation type, preventing the pretrained model from learning task-specific biases.

3.3. Architecture

The following architecture remains *strictly identical* across θ_0 pre-training and all downstream fine-tuning tasks. The model relies on a conditional U-Net architecture (*Improve-dUNet*) that processes a 6-channel tensor obtained by concatenating the noisy latent x_t and the spatial conditioning map. The network features an initial feature extraction block followed by 4 downsampling and 4 upsampling stages. Downsampling is performed via strided convolutions (4×4 , stride 2), which progressively double the channel width at each stage ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$).

Each architectural stage is built around residual *UNet-Blocks* comprising two convolutional layers (3×3), Group Normalization (4 groups), and SiLU activations. Timestep embedding is achieved using a Sinusoidal projection layer ($C = 64$) followed by a two-layer multi-layer perceptron with GELU activation to output a 256-dimensional time embedding vector, which is linearly projected and added to the intermediate features of every residual block.

Multi-head self-attention (2 heads with Layer Normalization) is applied at the deepest encoder stage, the bottleneck, and the first decoder stage. The decoder upsamples features using transposed convolutions with skip connections,

and a final convolution projects the 64-channel output to a 3-channel v -prediction.

3.4. [TODO: find a title]

Each specialist reuses θ_0 's architecture and the v -prediction framework, incorporating a cosine noise schedule over $T = 200$ diffusion steps, an exponential moving average (EMA) with $\beta = 0.999$, and the AdamW optimizer. Fine-tuning is conducted exclusively on each specialist's respective (condition, target) pairs from Sec. 3.1. The optimization utilizes a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} , regulated by a cosine annealing learning rate scheduler over a total of 100 epochs.

Two-stage grid search. [TODO: fill this from the notebook, explain in detail how the two stage grid search works]

4. Results

4.1. Choosing the search objective

We first developed and validated the search objective on Task Arithmetic, iterating through several evaluation metrics; the best-performing configuration was then applied unchanged to TIES and DARE-TIES on the final iteration to maximise the results.

Raw, pooled MSE. The earliest attempts scored candidate merges by MSE against ground truth, first at a single hand-picked λ point and then over an actual grid, split into a two-stage. However both consistently picked $\lambda_{SU} = 0$: because the super-resolution folder carries by far the largest absolute MSE, any pooled objective is minimised by *suppressing* that task rather than balancing it.

Normalised MSE. We proceeded to measure each task's error relative to its own single-task expert, alongside PSNR/SSIM for a fuller picture. This normalised objective was the first to select a genuine three-task compromise instead of switching a task off entirely. Combined-degradation folders have no single expert to normalise against, so they were instead scored relative to θ_0 — i.e. how much merging improves on doing nothing.

Perceptual Z-score. The final refinement replaced MSE with LPIPS, a perceptual metric, in order to compensate any mismatch between low MSE and visibly poor super-resolution output quality. More specifically, LPIPS was used as a replacement for MSE on any folder involving super-resolution, and put every folder on a common scale via a per-folder Z-score before ranking — $z = (x - \mu)/\sigma$ across the candidate grid, averaged before ranking.

[TODO: make this non-redundant and without repetitions]

[TODO: in sta tabella vanno accchittati i numeri perché non va bene + centra le robe]

Selection metric	λ grid	Winner
Raw MSE, single point	1 pt	0.9/0.1/0.0
Raw MSE, two-stage grid	27 combos	1.0/0.1/0.0
Normalised MSE (proposed)	≤ 4 combos	truncated run
Normalised MSE, two-stage, bug fixed	36 combos	0.9/0.1/0.1
Perceptual (LPIPS + MSE) Z-score	8 combos	0.9/0.1/0.1

Table 1. Evolution of the search objective. Values of λ reported as (SR/IR/SU).

[TODO: add a sentence like "however, when troubleshooting..."] The merge has the *lowest* MSE of the three, yet by far the *worst* LPIPS: MSE was rewarding a smoother, perceptually worse prediction. The LPIPS gap persists even with many more sampling steps, ruling out a sampling artifact. The task vectors are nearly orthogonal (cosine similarity ≈ 0.08 – 0.12), which normally favours task arithmetic, but τ_{SU} 's much larger norm means even a small mixing coefficient from the other two tasks perturbs its direction enough to visibly degrade perceptual quality. [TODO: make this paragraph shorter]

Findings The whole arc reduces to one lesson: the choice of evaluation metric mattered *far more* than the choice of search algorithm. Raw pooled MSE quietly answered a different question — “minimise total pixel error” — than the one actually being asked — “be good at all three tasks at once” — which is why it kept switching super-resolution off. Normalising against per-task baselines fixed that pooling artifact and recovered a real compromise, but only a perceptual metric exposed the deeper issue: interference between task vectors that pixel-wise error cannot see. Any merge of these experts should be validated perceptually before being trusted. [TODO: make this paragraph more paper-like, and also shorter]

[TODO: add caption of the following] VAL LOSS: MSE sulla predizione di v (v -prediction) a un timestep casuale del processo diffusivo (non è una metrica nello spazio immagine)

[TODO: add the following caption] VAL MSE: è la MSE nello spazio immagine (range $[-1,1]$) tra l'immagine generata con sampling DDIM completo e il target reale

[TODO: add the following caption] VAL PSNR

Eval. folder	Task Arith.	TIES	DARE-TIES
SR only	3.01e-3	3.37e-3	7.31e-3
IR only	6.80e-4	8.33e-4	1.13e-3
SU only	5.25e-2	5.20e-2	2.76e-2
SR+IR	1.36e-3	1.63e-3	2.09e-3
SR+SU	9.31e-3	9.36e-3	2.84e-2
IR+SU	1.49e-1	1.52e-1	6.47e-2
SR+IR+SU	5.60e-2	5.61e-2	2.67e-2

Eval. folder	Task Arith.	TIES	DARE-TIES
SR only	1.05e-3	1.14e-3	1.25e-3
IR only	3.62e-4	4.21e-4	4.56e-4
SU only	7.73e-3	7.76e-3	8.56e-3
SR+IR	5.14e-4	5.73e-4	6.18e-4
SR+SU	5.71e-3	5.71e-3	6.14e-3
IR+SU	1.62e-3	1.67e-3	1.69e-3
SR+IR+SU	2.65e-3	2.67e-3	2.68e-3

[TODO: can you make a graph/histogram(s) of these tables? for example some visualization that incorporates these tables]

4.2. Effect of post-merge fine-tuning

Applied zero-shot, all three merging methods produced visible artifacts and degraded reconstruction fidelity, confirming the destructive interference discussed above. We therefore fine-tuned each merged checkpoint with a short, deliberately conservative schedule that initializes *both* the network and its EMA copy from the merged weights and trains on a small, balanced mixture of all seven degradation combinations (IR, SR, SU and their pairings and triple). The formulation is kept identical to the specialists

Findings. The effect is consistent across all three merging methods: the brief fine-tune resolves the interference introduced by merging and recovers multi-task quality that the zero-shot merges never reach. Task Arithmetic, TIES, and DARE-TIES behave essentially identically here – each yields the same large, stable improvement over its own un-fine-tuned counterpart across every evaluation folder, in both pixel-level (MSE) and perceptual (LPIPS) terms, with no task collapsing.

5. Discussion and Conclusions

[TODO: add in some way or another this sentence in the conclusion] In other words, the post-merge step is what makes weight-space merging viable for these generative restoration tasks in the first place, and the specific merging algorithm matters far less than whether this stage is applied at all.

[TODO: write this section, based on the rest of the

Eval. folder	Task Arith.	TIES	DARE-TIES
SR only	39.29	38.63	37.85
IR only	41.54	40.70	40.29
SU only	33.62	33.28	32.96
SR+IR	39.42	38.85	38.32
SR+SU	31.53	31.49	31.19
IR+SU	35.48	35.31	35.28
SR+IR+SU	32.79	32.70	32.75

work, as best as you can. ASSUME THAT THERE IS TABLE WHICH SHOWS THAT Z SCORES IMPROVE IN THE FINE TUNED MERGED MODELS]

6. Statement on the use of AI

Three AI tools were used in this project, in distinct roles:

[TODO: write this list better]

- *Claude* was used for code generation, architectural choices, troubleshooting during training and generation of this report
- *Gemini* was used strictly for code generation
- *ChatGPT* was used for the writing of this report

All reported architectural choices, training curves, and quantitative results were produced and verified by the authors.

References

- Ho, J. and Salimans, T. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.
- Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 6840–6851, 2020.
- Ilharco, G., Ribeiro, M. T., Wortsman, M., Schmidt, L., and Hajishirzi, H. Editing models with task arithmetic. In *International Conference on Learning Representations (ICLR)*, 2023.
- Misiura, N. Starnet: Star removal software for astronomical images. <https://www.starnetastro.com/>, 2020. Accessed: 2026-07-09.
- Nichol, A. Q. and Dhariwal, P. Improved denoising diffusion probabilistic models. In *International Conference on Machine Learning (ICML)*, pp. 8162–8171, 2021.
- Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation.

In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241, 2015.

Salimans, T. and Ho, J. Progressive distillation for fast sampling of diffusion models. In *International Conference on Learning Representations (ICLR)*, 2022.

Space Telescope Science Institute (STScI). Hubble legacy archive (hla). <https://hla.stsci.edu/>, 2006. STScI, ESA, and CADAC.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Yadav, P., Albahahar, D., Choshen, L., Raffel, C., and Bansal, M. Ties-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pp. 7175–7193, 2023.

Yu, L., Yu, B., Yu, H. J., Li, J., Liu, Z., Wang, D., Zhang, F., and Lin, W. Language models are super mario: Absorbing abilities from homologous models through a task vector archive. In *International Conference on Learning Representations (ICLR)*, 2024.

Appendix

[TODO: Optional: move here anything cut from the main 2 pages, e.g. the full grid-search table, extra qualitative samples, the exact degradation-pipeline hyperparameters once finalized, or a derivation of the TIES trim/elect/merge steps if you want to show you understand them in detail.]